1: Ethical Business Plan

1.A. Company Name: WALLE-AI

1.B. Long-Term Vision Statement

- **1.B.1 Goals:**

  WALLE-AI aims to develop intelligent AI-powered robots that assist people with everyday tasks, with a strong focus on helping elderly and disabled individuals live more independently. The company's goal is to improve quality of life by making daily activities easier, safer, and more efficient while also reducing the burden on caregivers and families. In the long term, Walle-AI seeks to become a leader in assistive robotics by creating scalable, affordable solutions that can be used in homes, care facilities, and communities around the world.

- **1.B.2 Idea Origination:**

  The idea for Walle-AI originated from observing how many elderly and disabled individuals struggle with basic daily tasks such as cleaning, organizing, remembering important routines, and navigating tasks out in the real world. This need became even more clear through real-life experiences, including one of our groupmates working at In-N-Out and noticing that elderly customers often need assistance with simple actions like opening doors. These observations, along with discussions about technology and society and the rapid growth of AI capabilities, highlighted an opportunity to create meaningful solutions. Seeing how robotics could be used not just for convenience but to solve real human challenges inspired the creation of a company focused on using AI to directly improve people's everyday lives.

- **1.B.3 Purpose/Values/Mission:**

  WALLE-AI is a socially assistive artificial intelligence platform designed to enhance community engagement for elderly and disabled individuals. The system operates through a mobile application and optional in-home smart display device that provides personalized event reminders, conversational companionship, and volunteer coordination services. WALLE-AI uses natural language processing and behavioral analytics to identify opportunities for users to participate in community activities such as senior center events, religious gatherings, cultural programs, medical support groups, and volunteer opportunities.

  Key stakeholders include elderly users (typically age 65+), disabled adults of varying ages, family caregivers, local community organizations, healthcare providers, volunteer

groups, and the company operating WALLE-AI. Most AI computation occurs securely on encrypted cloud servers, while limited personalization data and reminders are cached locally on user devices. Behavioral interaction data and survey responses are encrypted and stored in company-managed databases. Limited reports may be shared with healthcare providers or families only with user consent.

- **1.B.4 Key Questions:**

  The key questions guiding Walle-AI include: How can we design robots that truly improve people's daily lives without replacing important human interaction? How can we make advanced AI technology affordable and accessible to those who need it most? How can we ensure that our robots operate safely, responsibly, and in a way that builds trust with users?

1.C. Strategy with Ethical Impacts AND Ethical Safeguards:

OKR1:

1.C.1.1:

WALLE-AI aims to increase quality of life for elderly and disabled individuals by automating social coordination and daily engagement tasks. The platform reduces reliance on external assistance by identifying community events, scheduling participation, sending reminders, and coordinating volunteers or transportation. The system adapts to users' communication abilities and mobility levels through natural language processing and behavioral analytics. Key stakeholders include elderly and disabled users, caregivers, assistive service organizations, and healthcare providers.

Key Results:
- 35% reduction in reliance on external assistance
- 90% successful task execution rate
- 80% improvement in ease of living (observer-validated)
- 20% increase in employment or structured community participation

1.C.1.2:

To ensure accuracy and reduce bias, data will be primarily collected through trained observers and system-logged behavioral data rather than self-reported user feedback. Observers may include caregivers, researchers, or trained study staff who document user interactions and outcomes in real-world or controlled settings.

External Assistance Reduction Rate (EARR)
EARR = (Baseline assistance − Post-use assistance) / Baseline assistance × 100
Success: ≥35% reduction after 3 months

Data Collection Method:
Observers (caregivers or study staff) will log all assistance events during both the 4-week baseline period and the 12-week WALLE-AI usage period. Assistance events include physical help, scheduling support, or task intervention. Logs will be standardized and time-stamped to ensure consistency.

Task Execution Success Rate (TESR)
TESR = Successful automated tasks / Total attempted tasks × 100
Success: ≥90%

Data Collection Method:
Task outcomes will be captured through a combination of automated system logs and observer verification. A task is considered successful if it is completed independently, on time, and without human override. Observers will validate ambiguous cases (e.g., partial completion or delayed execution).

Employment Participation Rate (EPR)
EPR = Employed users / Total participants × 100
Success: 20% increase within 12 months

Data Collection Method:
Employment status will be verified through observer check-ins and third-party confirmation where possible (e.g., caregiver reports, program participation records, or employer verification when available), rather than relying solely on user self-reporting. Observers will conduct quarterly structured interviews with both users and caregivers to confirm changes.

Ease of Living Score (ELS)

ELS = Composite observer-rated quality of life index

Success: 80% improvement from baseline

Data Collection Method:
Instead of relying on self-reported surveys alone, ELS will be measured using a standardized observational rubric completed by trained evaluators. The rubric will assess:
- Task independence
- Frequency of assistance required
- Signs of frustration or difficulty

- Routine consistency
- Engagement in daily activities

Observers will conduct scheduled in-home or simulated environment evaluations and assign scores based on consistent criteria. Limited user feedback may be collected for supplementary insight, but it will not be the primary metric.

1.C.1.3:

Key risks include caregiver job displacement, economic inequality, AI bias, and dependency risk. Frey and Osborne estimate that "about 47 percent of total US employment is at risk" from automation [1], highlighting workforce disruption concerns. AI bias may cause unequal performance across age or disability groups, reflecting disparities documented in prior AI systems. Overreliance on automation may reduce autonomy, contradicting ACM guidance to avoid harm and respect human agency [2].

Ethical Risk Table

Stakeholder	**Financial**	**Privacy**	**Conflict**	**Rights**
Disabled Users	mid	mid	mid	low
Caregivers	high	low	high	low
Companies	high	mid	high	mid
Healthcare Providers	low	mid	mid	low

Users face moderate privacy and financial risk due to data collection and affordability. Caregivers face high financial risk from displacement. The company carries high financial and conflict risk due to liability and performance pressure.

1.C.1.4:

WALLE-AI will implement:
• Quarterly bias audits ensuring <5% performance disparity across groups
• Transparent AI limitations through onboarding education and annual reports
• Caregiver retraining programs to mitigate displacement
• Data minimization, encryption, and privacy impact assessments

Effectiveness will be measured via observer-based equity audits, bias gap metrics, employment transition tracking, and privacy satisfaction assessments collected through structured interviews rather than passive self-reporting alone.

OKR2:

1.C.2.1:

Objective: Increase Community Integration for Elderly and Disabled Users through AI-assisted engagement

WALLE-AI improves community integration by actively coordinating real-world participation in events, volunteer opportunities, and social activities. The system not only recommends activities but also assists with scheduling, reminders, and transportation coordination to ensure successful participation.

Key Results:
• 25% increase in verified community participation within 6 months
• 85% successful AI-assisted engagement completion rate
• 80% adoption rate of WALLE-AI community features
• 20% increase in successful volunteer coordination matches

Primary stakeholders include elderly and disabled users, caregivers, community organizations, healthcare providers, and local governments.

1.C.2.2:

To ensure accuracy, data will be collected through a combination of system logs and trained observers, rather than relying on self-reported user data.

Community Participation Rate (CPR)
CPR = Verified attended events / Total recommended events × 100
Success: 25% increase from baseline within 6 months

Data Collection Method:
• Who tracks: Caregivers, community staff, and trained research observers
• How: Attendance is verified through check-ins (QR codes, staff logs, or observer

confirmation)
• When:

- 30-day baseline period (no WALLE-AI assistance)
- 6-month tracking period with WALLE-AI
  • Observers record whether attendance was independent, AI-assisted, or required external intervention

AI-Assisted Engagement Success Rate (AESR)
AESR = Successful AI-assisted engagements / Total AI-initiated engagements × 100
Success: ≥85%

Definition of Success:
An engagement is successful if the user:
• Receives the recommendation
• Completes scheduling
• Attends or participates without failure or cancellation

Data Collection Method:
• Who tracks: Automated system logs + observer verification
• How:

- System logs track recommendations, reminders, and scheduling
- Observers validate completion and flag failures (missed events, confusion, cancellations)
  • When: Continuous logging over 6 months with biweekly observer review sessions

Feature Adoption Rate (FAR)
FAR = Active users of community features / Total users × 100
Success: ≥80%

Data Collection Method:
• Who tracks: System analytics team
• How: Backend logs track usage of features (event browsing, reminders, volunteer coordination)
• When: Weekly usage summaries over a 6-month period
• Adoption requires consistent use (≥3 interactions per week), not just initial setup

Volunteer Coordination Success Rate (VCSR)
VCSR = Successful matches / Total coordination attempts × 100
Success: ≥20% increase

Data Collection Method:
• Who tracks: Community partners + observers
• How:

- Matches are logged by the system
- Success is confirmed through follow-up by observers or partner organizations
  • When: Monthly reporting cycles with cross-verification from partner organizations

WALLE-AI Functional Engagement Assessment (WFEA)

To replace indirect measures like loneliness, WALLE-AI will use a direct observational assessment of real-world engagement ability.

WFEA Score Components:
Observers evaluate users across 5 categories:
• Ability to understand and act on recommendations
• Independence in attending events
• Consistency of participation
• Need for external assistance
• Responsiveness to reminders

Scoring:
Each category rated 1–5 → Total score converted to percentage

Success: ≥30% improvement from baseline

Data Collection Method:
• Who tracks: Trained evaluators and caregivers
• How: Structured observation during real-world or simulated engagement tasks
• When:

- Baseline (pre-adoption)
- 3-month evaluation
- 6-month evaluation

This ensures measurement is directly tied to WALLE-AI's core goal: improving real-world participation, not abstract emotional states.

1.C.2.3:

> A major ethical concern is emotional manipulation and overdependence. If AI conversational features simulate emotional closeness, users may substitute artificial interaction for real relationships. Additionally, optimizing for engagement metrics may

conflict with user autonomy if the system prioritizes retention over meaningful participation.

Expected Ethical Impact Risk Table:

Stakeholder	Financial	Privacy	Conflicting Interest	Violation of Rights
Elderly/ Disabled User	Low	High	Mid	Mid
Family Members	Low	Mid	Mid	Low
Company	High	Mid	High	Mid
Community Organizations	Low	Low	Mid	Low
Government/ Regulators	Mid	Low	Mid	Low

Users face high privacy risk due to behavioral tracking. The company faces high financial and conflicting-interest risks if lawsuits or regulatory penalties arise.

1.C.2.4:

To mitigate these risks, WALLE-AI will:
- Clearly disclose AI identity during all interactions
- Require explicit opt-in for behavioral tracking
- Cap reminder frequency to prevent overdependence
- Establish a multidisciplinary ethics board to review engagement algorithms quarterly

Effectiveness Measurement:
- Observer-based audits of engagement authenticity
- Tracking ratio of real-world vs AI-only interaction
- Complaint logs and user interviews (conducted by researchers, not passive surveys)
- Third-party audits of algorithmic behavior

OKR3:

1.C.3.1:

Objective: Ensure WALLE-AI is financially accessible and equitably distributed across different income groups

Definitions:
Affordable: A product is considered affordable if the total monthly cost of WALLE-AI (device + subscription) does not exceed 5% of a household's monthly income.

Equitable: Access to WALLE-AI is considered equitable if adoption rates and usability outcomes do not significantly differ (≤10% disparity) across income groups, racial groups, and geographic regions, after controlling for access factors such as internet availability.

WALLE-AI will prioritize income-based pricing models, subsidies, and accessibility programs to ensure that low-income users can benefit at rates comparable to higher-income users.

Key Results:
- 70% of users across all income brackets meet the affordability threshold (≤5% income spent)
- ≤10% disparity in adoption rates between low-income and high-income users
- 3+ government or nonprofit subsidy partnerships targeting low-income populations
- 85% retention rate across all income groups

Primary stakeholders include elderly and disabled users (especially low-income households), caregivers, healthcare providers, nonprofit organizations, and government agencies.

1.C.3.2:

To accurately measure affordability and equity, WALLE-AI will rely on income-verified data, structured surveys, and observer-supported validation, rather than assumptions about cost reductions alone.

Affordability Ratio (AR)

AR = Monthly WALLE-AI Cost / Monthly Household Income

Success: ≥70% of users have AR ≤ 5%

Data Collection Method:
• Who tracks: Research team + financial verification staff
• How:

- Users provide income ranges (not exact income) through a structured survey
- Cost data is automatically tracked
- Observers or staff assist users (especially elderly/disabled) in accurately completing financial inputs
  • When:
- At onboarding
- Re-evaluated at 3 months and 12 months

Equity of Adoption Rate (EAR)

EAR = Adoption rate (low-income group) ÷ Adoption rate (high-income group)

Success: ≥0.90 (no more than 10% disparity)

Data Collection Method:
• Who tracks: Data analytics team + external auditors
• How:

- Users are grouped by income brackets
- Adoption is defined as consistent usage (≥3 times per day)
  • When: Monthly tracking over a 6–12 month period

Subsidized Access Rate (SAR)

SAR = Users receiving financial assistance / Total low-income users × 100

Success: ≥50% of eligible low-income users receive subsidies

Data Collection Method:
• Who tracks: Partnership coordinators + nonprofit/government partners
• How: Verified enrollment in subsidy programs
• When: Quarterly reporting

Retention Equity Rate (RER)

RER = Retention (low-income users) ÷ Retention (high-income users)

Success: ≥0.90

Data Collection Method:
- Who tracks: System analytics team
- How: Usage logs + follow-up interviews conducted by researchers
- When: Measured over a 12-month period

Income-Based Affordability Survey

To ensure affordability is grounded in real user financial situations, WALLE-AI will implement a structured affordability survey:

Administered by: Trained staff or caregivers (to avoid misunderstanding)
When: Onboarding, 3 months, 12 months

Survey Questions:

1. What is your household monthly income range?
   - <$1,500
   - $1,500–$3,000
   - $3,000–$5,000
   - $5,000+
2. What is your monthly spending on WALLE-AI (device payment + subscription)?
3. Does the cost of WALLE-AI feel manageable within your budget?
   - Yes, easily manageable
   - Manageable with some difficulty
   - Difficult to afford
   - Not affordable
4. Have you had to reduce spending on other essentials (food, medicine, utilities) to afford WALLE-AI?
   - Yes / No
5. Would financial assistance (subsidy, payment plan) significantly improve your ability to use WALLE-AI?
   - Yes / No
6. Are there any barriers preventing you from continued use?
   - Cost
   - Internet access
   - Technical difficulty
   - Other

Validation:
Responses will be cross-checked with actual usage and payment data, and observers will assist users to ensure accurate understanding.

1.C.3.3:

Key ethical concerns to making WALLE-AI affordable and equitable are the digital divide or technological inequality in low-income households. Even with WALLE-AI being more affordable, there are still factors that communities can lack such as internet access, or technical literacy which ends up causing WALLE-AI to benefit wealthier users more. A similar ethical concern was highlighted in *Epic Games Inc. v Apple Inc.,* where Apple's control over the App Store limited access to lower-cost alternatives, which shows how control over technology distribution can impact disadvantaged users [3]. This connects to how WALLE-AI, despite being more affordable can unintentionally favor users that have better access to the internet and resources.

Another ethical concern is bias in distribution from non-profits and government organizations. For the free trial and access to WALLE-AI the distributors may prioritize easier to reach populations or exclude minorities.

Stakeholders	**Financial risk**	**Privacy risk**	**Conflicting interest**	**Violation of Rights**
Disabled/Elderly users	mid	high	medium	medium
Caregivers	mid	medium	mid	low
Company	high	low	high	low

Healthcare Providers	low	high	mid	low

Disabled/Elderly: Financial risk is medium because even with reduced costs the price is still not too low. Privacy risk can be high due to cameras and sensors being used to monitor its surroundings. Conflicting interest is medium since the users want an affordable easy to use robot while the company wants to make profits and collect important data. Violation of rights can be medium if the user is unaware of how the robot functions.

Caregivers: Medium financial risk since they are responsible for purchasing the robot. Privacy risk is medium since caregivers may be recorded while the robot is functioning. Conflicting interest is medium because caregivers may want more monitoring for convenience, while the elderly user may want more privacy. Violation of rights is low since caregivers are secondary users and are mostly aware of how the system works and what it does.

Company: Financial risk is high because reducing prices can also lower the company's profit. Privacy risk is low because the company owns and controls how the data is stored. Conflicting interest is high because the company wants to maximize profits and data collection while customers want affordability and better privacy. Violation of rights is low since the company is following the law.

Healthcare Providers: Financial risk is low because the healthcare providers are not directly purchasing the robots with their money. Privacy risk is medium because they may be recorded while helping patients. Conflicting interest is medium because they may want to collect more data to understand patients while patients may want more privacy. Violation of rights is low since in a place with healthcare providers they must follow proper policies and practices.

1.C.3.4:

To address these concerns, WALLE-AI will:

- Implement income-based pricing tiers rather than uniform pricing
- Partner with internet providers to bundle low-cost connectivity
- Provide guided onboarding and training led by staff or caregivers
- Require partners to follow transparent, need-based distribution guidelines

Effectiveness Measurement:

- Affordability ratio trends across income groups
- Distribution audits to detect demographic gaps
- Observer-assisted usability assessments for low-income users
- Third-party equity audits to verify fair access

One method of solving the technological inequality and digital divide could be to include the product at a lower price when partnering with wifi companies. We could also give guided tutorials on how to use the robots. The company should be responsible for teaching customers how to use the robot. Through the use of a video or through a guided tutorial when purchasing. We can measure the effectiveness of this by asking if costumes understood how to use the robot after the guided tutorial or the video and see the difference.

To solve the other ethical concern of bias in distribution, WALLE-AI will implement clear and transparent guidelines for all governments and non-profit organizations where we require partners to distribute the robots equally and based on need rather than convenience ensuring that minorities are not excluded and some communities aren't prioritized over others. To see the effectiveness of this WALLE-AI will collect distribution data with the users consent including which communities received the robots and the demographic of the users[2]. This data can be used to inform WALLE-AI if any communities or minorities are not being prioritized.

References

[1] C. B. Frey and M. A. Osborne. 2017. *The future of employment: How susceptible are jobs to computerisation?* Technological Forecasting and Social Change 114, 254–280.

[2] Association for Computing Machinery (ACM). 2018. *ACM Code of Ethics and Professional Conduct.* ACM.

[3] Epic Games, Inc. v. Apple Inc., 67 F.4th 946 (9th Cir. 2023).F{}

[4] AI Now Institute. 2018. *AI Now Report 2018*. New York University.

[5] European Commission High-Level Expert Group on Artificial Intelligence. 2019. *Ethics Guidelines for Trustworthy AI*. European Commission.